PRANEETH REDDY NALAMALAPU

EMBEDDED SOFTWARE ENGINEER

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CAREER OBJECTIVE

Embedded Software Engineer with hands-on knowledge in C programming, Linux environment. Strong foundation in low-level programming, microcontrollers, and firmware development. Experienced in writing efficient embedded code, interfacing hardware peripherals, and debugging real-time applications. Focused on building reliable embedded solutions using C in resource-constrained environments. Looking for a challenging opportunity to grow in the electronics industry.

WORK EXPERIENCE

Advanced Embedded Systems Trainee
Emertxe Information Technologies, Bengaluru *(Present)*

- focusing on embedded software development, microcontrollers, embedded C programming, RTOS, Linux, and device driver fundamentals.
- Participating in hands-on projects involving firmware development, hardware interfacing, memory management, and real-time embedded applications.

Embedded Systems & IoT Intern
Pantech.AI | Oct 2025 – Feb 2026

- Programmed and interfaced **microcontrollers** with peripheral devices for real-time hardware control and monitoring.
- Integrated sensors and communication modules to support IoT-enabled embedded solutions.
- Applied embedded software development practices, including code debugging, hardware interfacing, and firmware optimization.

TECHNICAL SKILLS

Programming Languages:

- Advanced C programming
- Data Structures

System programming

- Linux Kernel system calls
- IPC mechanisms – Pipe, FIFO, Shared memory

Concepts

- Memory handling
- Interrupts
- GPIO
- Timers

Tools

- GCC
- Basic debugging tools

EDUCATION

- **BTech (EEE), QIS College of Engineering and Technology – Ongole, 6.77(cgpa) 2021 - 2025**
- **Class – XII, Sri Chaitanya JR College- Ongole, 546(marks), 2019 – 2021**
- **Class – X, Montessori high School- Ongole, 8.5(GPA), 2018 – 2019**

CONTRIBUT

- **LEETCODE: solved 50+ leetcode problems using c language**
  Link: https://leetcode.com/u/HO68lYDuGf

PROJECTS:

PROJECTS AT EMERTXE

Project 1:

Title: Image steganography using LSB Encoding and Decoding

The objective was to send a secret text file encoded inside an image of bmp file format. Encoded the length of the secret text and then encoded the data into the LSB of the image bytes. The decoding process involves decoding the length and then decoding the text bit by bit. The final output is the secret text after decoding.

Technologies used : File operations, Pointers, Bitwise operations, Functions, Makefiles, Command line arguments

Project 2:

Title: Address Book

Project Brief:

developed a command-line Address Book Management System to efficiently store and manage contact information. The application supports adding, searching, editing, deleting, and listing contacts while ensuring data persistence through file handling. It also incorporates input validation and sorting techniques to improve usability and data organization.

Technologies Used:

Structures, File Operations, Dynamic Memory Allocation, Pointers, Functions, String Handling, Sorting Algorithms, Command Line Interface.

Project 3:

Title: MP3 Tag Reader

MP3 tag reader is a software which will read and display MP3 (ID3) tag information from MP3 files. The software will be desktop based and not web based. This solution will read the given MP3 file, extract various tag information and display them via command line. This project can be extended to implement a tag editor, where-in users can modify mp3 tag information.

Technologies used – Function pointers – String operations – File I/O handling

Project 4:

Title: **Arbitrary Precision Calculator (APC)**

Project Brief:

Developed an arbitrary precision calculator capable of performing arithmetic, modulus, and power operations on large numbers beyond standard data type limits. Implemented linked-list-based data structures and optimized mathematical operations using ADT concepts.

Technologies used Order complexity – Applying Abstract Data Types (ADT) – Self-referential structures using linked list

ACADEMIC PROJECTS:

Project 3

Title: Automatic Crop Protection System

Project Brief:

Designed an embedded-based automated crop protection system using sensor inputs and motor control mechanisms. Implemented state-machine-based firmware for automatic detection and response, reducing manual intervention.

Technologies Used:

Embedded C – Microcontroller Programming, Sensor Interfacing, Motor Control, State Machine Design, GPIO, Embedded Firmware, Proteus Simulation.

Project 4

Title: Vehicle-to-Vehicle Energy Transfer System

Project Brief:

Developed an embedded control system for bidirectional vehicle-to-vehicle energy transfer using a DC-DC converter. Designed and simulated control algorithms for stable voltage regulation and efficient power management.

Technologies Used:

Embedded C – Microcontroller Programming, Power Electronics, Bidirectional DC-DC Converter Control, Voltage Regulation, Control Algorithms, MATLAB/Simulink, Embedded System Simulation.